

Register	Type	Register Name
Charger Control Message		
10001	RO	The type of machine
10002	WR	Serial number High
10003	WR	Serial number Low
10004	RO	Hardware version
10005	RO	Software version
10006	WR	PV voltage calibration coefficient
10007	WR	Battery voltage calibration coefficient
10008	WR	Charger current calibration coefficient
10009-10100	WR	reserved
10101	WR	charger work enable
10102	WR	Absorb voltage
10103	WR	Refloat voltage
10104	WR	Float voltage
10105	WR	Battery low voltage
10106	WR	reserved
10107	WR	Battery High voltage
10108	WR	Max charger current
10109	WR	Absorb charger current
10110	WR	Battery type
10111	WR	Battery AH
10112	WR	Remove the accumulated data

Charger Display Message

15201	RO	Charger workstate
15202	RO	Mppt state
15203	RO	charging state
15204	RO	reserved
15205	RO	PV voltage
15206	RO	Battery voltage
15207	RO	Charger current
15208	RO	Charger power
15209	RO	Radiator temperature
15210	RO	External temperature
15211	RO	Battery Relay
15212	RO	PV Relay
15213	RO	Error message
15214	RO	Warning message
15215	RO	BattVol Grade
15216	RO	Rated Current
15217	RO	Accumulated power high
15218	RO	Accumulated power low
15219	RO	Accumulated day
15220	RO	Accumulated hour
15221	RO	Accumulated minute

	Effective Range
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PC1600	
1.0.00	
1.0.00	
16384	
16384	
16384	
reserved	
0 : OFF 1 : ON	effective range : 0,1 the default value is 1 ;
0.1V	480~540(48.0-54.0)V the default value is 50.0V
0.1V	510~580(51.0-58.0)V the default value is 54.8V
0.1V	530~580(53.0-58.0)V the default value is 57.2V
0.1V	340~440(34.0-44.0)V the default value is 34.0V
0.1V	580~600(58.0-60.0)V the default value is 60.0V
0.1A	0.1A effective range : (0-80)A
0.1A	0.1A effective range : (0-80)A
0: no choose 1 : lead acid battery 2 : lithium battery	the default value is 0 ; effective range : 0,2
1AH	effective range : (0-900)AH the default value is 100AH
0 : No remove the accumulated data 1 : Remove the accumulated data	the default value is 0 ; effective range : 0,1

0 : Initialization mode 1 : Selftest Mode 2: Work Mode 3 : Stop Mode	
0 : Stop 1 : MPPT 2: Current limiting	
0 : Stop 1 : Absorb charge 2: Float charge	
0.1V	(0.0-150.0)V
0.1V	(0.0-80.0)V
0.1A	(0.0-90.0)A
1W	(0-5000)W
1°C	(-40-150)°C
1°C	(-40-150)°C
	0: Disconnect 1:Connect
	0: Disconnect 1:Connect
Refer to frame Charger Error message 1	
Refer to frame Charger Warning message 1	
1V	
0.1A	
1000KWH	
0.1KWH	
1day	
1hour	
1minute	

Register	Type	Register Name
Inverter Control Message		
20001	RO	The type of machine
20002	RW	Serial number High
20003	RW	Serial number Low
20004	RO	Hardware version
20005	RO	Software version
20006-20008	RW	reserved
20009	RW	Battery voltage calibration coefficient
20010	RW	Inverter voltage calibration coefficient
20011	RW	Grid voltage calibration coefficient
20012	RW	Bus voltage calibration coefficient
20013	RW	Control Current calibration coefficient
20014	RW	Inverter Current calibration coefficient
20015	RW	Grid Current calibration coefficient
20016	RW	Load Current calibration coefficient
20017-20100	RW	reserved
20101	RW	Inverter offgrid work enable
20102	RW	Inverter output voltage Set
20103	RW	Inverter output frequency Set
20104	RW	Inverter search mode enable
20105	RW	Inverter ongrid work enable
20106	RW	Inverter charger from grid enable
20107	RW	Inverter discharger enable
20108	RW	Inverter discharger to grid enable

20109	RW	Energy use mode
20110	RW	Time mode use enable
20111	RW	Grid protect standard
20112	RW	SolarUse Aim
20113	RW	Inverter max discharger current
20114	RW	Inverter max charger current
20115	RW	Grid max output current
20116	RW	Grid max input current
20117	RW	start charger voltage point
20118	RW	normal voltage point
20119	RW	start sell voltage point
20120	RW	Batt low protection mode
20121	RW	reserved
20122	RW	reserved
20123	RW	reserved
20124	RW	reserved
20125	RW	Grid max charger current set
20126	RW	Absorb charger current set
20127	RW	Battery low voltage
20128	RW	Battery high voltage
20129	RW	Battery type
20130	RW	Battery AH
20131	RW	battery low return voltage
20132	RW	Max Combine charger current
20133	RW	Charger Start Time (Hour)
20134	RW	Charger Start Time (Minutes)
20135	RW	Charger End Time (Hour)
20136	RW	Charger End Time (Minutes)
20137	RW	Discharger Start Time (Hour)

20138	RW	Discharger Start Time (Minutes)
20139	RW	Discharger End Time (Hour)
20140	RW	Discharger End Time (Minutes)
20142	RW	System setting
20143	RW	Charger source priority
20161	RW	Gridtie Safety type
20162	RW	Auto Test Start Cmd
20163	RW	
20164	RW	
20165	RW	Grid LVP Slow Protetion Value
20166	RW	Grid LVP Slow Protetion Time
20167	RW	Grid OVP Slow Protetion Value
20168	RW	Grid OVP Slow Protetion Time
20169	RW	Grid LVP Fast Protetion Value
20170	RW	Grid LVP Fast Protetion Time
20171	RW	Grid OVP Fast Protetion Value
20172	RW	Grid OVP Fast Protetion Time
20173	RW	Frequency low Protection Value
20174	RW	Frequency low Protection Time
20175	RW	Frequency High Protection Value
20176	RW	Frequency High Protection Time
20177	RW	
20178	RW	
20179	RW	
20180	RW	
20181	RW	
20182	RW	
20183	RW	Reconnect Time
20184	RW	
20185	RW	Q-U Characteristic Curve Start point
20186	RW	PF-P Characteristic Curve Power point
20187	RW	PF-P Characteristic Curve Volt. point
20188	RW	Power recover speed %Pn/min
20189-20190	RW	保留

20191	RW	Output Q Mode
20192	RW	Active Power Setting
20193	RW	Reactive Power Setting
20194	RW	
20195-20200	RW	保留
20201	RW	Year
20202	RW	Month
20203	RW	Day
20204	RW	
20205	RW	Hour
20206	RW	Minutes
20207	RW	Seconds
20208	RW	
20209	RW	Force charger enable
20210	RW	Force discharger enable
20211	RW	
20212	RW	
20213	RW	Remove the accumulated data
20214	RW	Reset the parameter

Inverter Display Message

25201	RO	work state
25202	RO	AC voltage grade
25203	RO	Rated power
25204	RO	reserved
25205	RO	Battery voltage
25206	RO	Inverter voltage
25207	RO	Grid voltage
25208	RO	BUS voltage
25209	RO	Control current
25210	RO	Inverter current
25211	RO	Grid current
25212	RO	Load current
25213	RO	PInverter
25214	RO	PGrid
25215	RO	PLoad
25216	RO	Load percent
25217	RO	SInverter
25218	RO	SGrid
25219	RO	Sload
25220	RO	reserved

25221	RO	Qinverter
25222	RO	Qgrid
25223	RO	Qload
25224	RO	reserved
25225	RO	Inverter frequency
25226	RO	Grid frequency
25227	RO	reserved
25228	RO	reserved
25229	RO	Inverter max number
25230	RO	Combine type
25231	RO	Inverter number
25232	RO	reserved
25233	RO	AC radiator temperature
25234	RO	Transformer temperature
25235	RO	DC radiator temperature
25236	RO	reserved
25237	RO	Inverter relay state
25238	RO	Grid relay state
25239	RO	Load relay state
25240	RO	N_Line relay state
25241	RO	DC relay state
25242	RO	Earth relay state
25243	RO	reserved
25244	RO	reserved
25245	RO	Accumulated charger power high
25246	RO	Accumulated charger power low
25247	RO	Accumulated discharger power high
25248	RO	Accumulated discharger power low
25249	RO	Accumulated buy power high
25250	RO	Accumulated buy power low
25251	RO	Accumulated sell power high
25252	RO	Accumulated sell power low
25253	RO	Accumulated load power high
25254	RO	Accumulated load power low
25255	RO	Accumulated self_use power high
25256	RO	Accumulated self_use power low
25257	RO	Accumulated PV_sell power high
25258	RO	Accumulated PV_sell power low
25259	RO	Accumulated grid_charger power high
25260	RO	Accumulated grid_charger power low

25261	RO	Error message 1
25262	RO	Error message 2
25263	RO	Error message 3
25264	RO	reserved
25265	RO	Warning message 1
25266	RO	Warning message 2
25267	RO	reserved
25268	RO	reserved
25269	RO	Serial number High
25270	RO	Serial number Low
25271	RO	Hardware version
25272	RO	Software version
25273	RO	Batt power
25274	RO	Batt current

Physical Unit	Effective Range
PH1000/PH3000/PH1800	
1.00.00	
1.00.00	
reserved	
16384	
16384	
16384	
16384	
16384	
16384	
16384	
16384	
16384	
reserved	
0 : OFF 1 : ON	effective range : 0,1 the default value is 1 ;
220.0V-240.0V	(2200-2400)
50.00Hz/60.00Hz	5000/6000
0 : OFF 1 : ON	effective range : 0,1 the default value is 0 ;
0 : OFF 1 : ON	effective range : 0,1 the default value is 1 ;
0 : OFF 1 : ON	effective range : 0,1 the default value is 1 ;
0 : OFF 1 : ON	effective range : 0,1 the default value is 1 ;
0 : OFF 1 : ON	effective range : 0,1 the default value is 1 ;

0:PL ; 1:FL; 2:FS; 3:UPS; 4: PO(Old) 0:SUB;1SBU;3:UTI;(New)	
0 : disable 1 : enable	
0 : VDE4105; 1 : UPS ; 2 : home	
0:LBU 1:BLU(defalut)	
0.1A (AC)	1.0A-17.4A (10-174)
0.1A (AC)	1.0A-17.4A (10-174)
0.1A (AC)	0A-26.0A (0-260)
0.1A (AC)	0A-26.0A (0-260)
0.1V	450~540(45.0-54.0)V the default value is 48.0V
0.1V	480~550(48.0-55.0)V the default value is 53.0V
0.1V	500~540(50.0-58.0)V the default value is 54.0V
0:usually_defined 1:user_defined	
reserved	
reserved	
reserved	
reserved	
0.1A(DC)	10~800(1.0-80.0)A the default value is 60.0A
0.1A(DC)	10~300(1.0-30.0)A the default value is 10.0A
0.1V	400~440(40.0-44.0)V the default value is 42.0V
0.1V	580~600(58.0-60.0)V the default value is 60.0V
0: no choose 1:Use_defined battery 2:lithium battery 3:Sealed_lead battery 4:AGM battery (default) 5:GEL battery 6:Flooded battery	the default value is 4; effective range : 0,6
1AH	effective range : (0-900)AH the default value is 100AH
0.1V	400~500(40.0-50.0)V the default value is 44.0V
0.1A(DC)	10~1400(1.0-140.0)A the default value is 60.0A
1h	
1min	
1h	
1min	
1h	

1min	
1h	
1min	
	refer to the frame System setting bit
0:Solar first 2:Solar and Utility(default) 3:Only Solar	
0 : VDE4105 1 : AS4777 2 : CEI021 3 : user define	
1 : Start 0 : Stop	
0.1V	LVP:Low voltage protection
10ms	
0.1V	OVP:Over voltage protection
10ms	
0.1V	
10ms	
0.1V	
10ms	
0.01Hz	
10ms	
0.01Hz	
10ms	
1s	
0.01v	
1%	
0.1v	
1%	

0 : PF Curv, 1: QU Curv	
w	
w	
0 : Enable 1 : Disable	
0 : Enable 1 : Disable	
0 : No remove the accumulated data 1 : Remove the accumulated data	the default value is 0; effective range:0,1
0 : No effect 1 : Action	the default value is 0; effective range:0,1

0:PowerOn 1:SelfTest 2:OffGrid 3:Grid-Tie 4:ByPass 5:Stop	
230:230V 120:120V	
PV1800 VA/PH3000 W	
0.1V	
0.1V	
0.1V	
0.1V	
0.1A	
0.1A	
0.1A	
0.1A	
1W	
1W	
1W	
1%	new
1VA	
1VA	
1VA	

[illegible]

	Refer to frame Error message 1
	Refer to frame Error message 2
	Refer to frame Error message 3
	Refer to frame Warning message 1
	Refer to frame Warning message 2
1.00.00	
1.00.00	
W	
A	

note	
1:Turn on the inverter output when the grid is off. output when the grid is off.	0:Shut down the inverter
Set the output voltage amplitude.	
Set the output frequency.	
0:If disabled, no matter connect load is low or high, the on/off status of inverter output will not be effected. 1:If enable, the inverter begins search mode if the AC load connected is pretty low or not detected. The inverter's "search" mode reduces stand-by energy consumption during no-load conditions.	
1:Enable the inverter connect to an AC input source. connect to an AC input source.	0:Disable the inverter
1:Enable grid charge	0:Disable grid charge
1:Enable Inveter discharging from battery when the Inverter is connnected to Grid 0:Disable Inveter discharging from battery when the Inverter is connnected to Grid	
1:Enable Inveter discharging from battery to grid when the Inverter is connnected to Grid 0:Disable Inveter discharging from battery to grid when the Inverter is connnected to Grid	

[illegible]

[illegible]

[illegible][illegible]

[illegible]

[illegible]

System setting bit

Bit	meaning
0	OverLoadRestartForbid
1	OverTempRestartForbid
2	OverLoadBypassForbid
3	AutoTurnPageFlagForbid
4	GridBuzzEnable(only use by PV1800)
5	BuzzForbide(only use by PV1800)
6	LcdLightEnable
7	RecordFaultForbid
8	reserved
9	reserved
10	reserved
11	reserved
12	reserved
13	reserved
14	reserved
15	reserved

Inverter error message 1

Bit	meaning	Bit
0	Fan is locked when inverter is off	0
1	Inverter transformer over temperature	1
2	battery voltage is too high	2
3	battery voltage is too low	3
4	Output short circuited	4
5	Inverter output voltage is high	5
6	Overload time out	6
7	Inverter bus voltage is too high	7
8	Bus soft start failed	8
9	Main relay failed	9
10	Inverter output voltage sensor error	10
11	Inverter grid voltage sensor error	11
12	Inverter output current sensor error	12
13	Inverter grid current sensor error	13
14	Inverter load current sensor error	14
15	Inverter grid over current error	15

Inverter error message 2

Bit	meaning	Bit
0	Inverter radiator over temperature	0
1	Solar charger battery voltage class error	1
2	Solar charger current sensor error	2
3	Solar charger current is uncontrollable	3
4	Inverter grid voltage is low	4
5	Inverter grid voltage is high	5
6	Inverter grid under frequency	6
7	Inverter grid over frequency	7
8	Inverter over current protection error	8
9	Inverter bus voltage is too low	9
10	Inverter soft start failed	10
11	Over DC voltage in AC output	11
12	Battery connection is open	12
13	Inverter control current sensor error	13
14	Inverter output voltage is too low	14
15	reserved	15

Inverter warning message 1

meaning

Fan is locked when inverter is on.

Fan2 is locked when inverter is on.

Battery is over-charged.

Low battery

Overload

Output power derating

Solar charger stops due to low battery.

Solar charger stops due to high PV voltage.

Solar charger stops due to over load.

Solar charger over temperature

PV charger communication error

reserved

reserved

reserved

reserved

reserved

Inverter warning message 2

meaning

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

Charger error message 1		Charger Warning n	
Bit	meaning	Bit	
0	Hardware protection	0	
1	Over current	1	
2	Current sensor error	2	
3	Over temperature	3	
4	PV voltage is too high	4	
5	PV voltage is too low	5	
6	Battery voltage is too high	6	
7	Battery voltage is too Low	7	
8	Current is uncontrollable	8	
9	Parameter error	9	
10	reserved	10	
11	reserved	11	
12	reserved	12	
13	reserved	13	
14	reserved	14	
15	reserved	15	

message 1

meaning

Fan Error

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved

reserved